Protocol for Enterokinase enzyme activity assay:

1. **Reagent/ Chemicals used:**

Table 1: Reagents and chemicals		
S. No.	**Components**	**Make**
1.	Enterokinase from porcne intestine	Sigma
2.	Tris base	Himedia
3.	Glycine	Himedia
4.	SDS	SRL
5.	Acrylamide: Bisacrylamide	NA
6.	Ammonium persulphate	SRL
7.	Coomassie brilliant blue	Sigma
8.	Glacial acetic acid	Emplura
9.	Methanol	RCI Labscan
10.	Glycerol	SRL
11.	Tricine	SRL
12.	Calcium chloride	SRL
13.	Sodium chloride	CDH
14.	TEMED	Sigma
15.	HCl	Merck
16.	Acrylamide	Merck
17.	Bisacrylamide	Sigma

2. **Equipments/Instruments used:**

Table 2: Equipments/ Instruments			
S. No:	**Name**	**Make**	**Equipment/Instrument ID**
1.	Thermomixer	Eppendorf	BG-TMC-003
2.	Adjustable mono-channel and multi-channel pipettes	Eppendorf or equivalent	BG-PIP-114, BG-PIP-158, BG-PIP-176, BG-PIP-083, BG-PIP-087, BG-PIP-091

3. **Reagent Preparations:**

- **Dilution/Storage Buffer:** To prepare 50 ml of dilution/storage buffer containing 20 mM Tris HCl, 200 mM NaCl, 2 mM $CaCl_2$, and 50% glycerol, dissolve 0.121 g of Tris HCl, 0.584 g of NaCl, and 0.011 g of calcium chloride in 35 ml of deionized water. Add 25 ml of glycerol, adjust the pH to 7.4, and make up the final volume to 50 ml with deionized water.
- **Cleavage/Capture Buffer:** To prepare 50 ml of cleavage/capture buffer containing 200 mM Tris HCl, 500 mM NaCl, and 20 mM $CaCl_2$, dissolve 1.211 g of Tris HCl, 1.461 g of NaCl,

and 0.111 g of calcium chloride in 35 ml of deionized water. Adjust the pH if necessary and make up the final volume to 50 ml with deionized water.

- **Ammonium persµlphate (APS) 14%:** Weigh 0.14 g of APS and dissolve in 1 mL of milli-Q or purified water and store the solution at -20°C.
- **Preparation of staining solution- 1000 mL:** To prepare 1000 mL of gel staining solution, add the following in a beaker and mix,

- Coomassie brilliant blue - 0.5 g
- Milli-Q or purified water - 450 mL
- Glacial acetic acid - 100 mL
- Methanol - 450 mL

- **Preparation of destaining solution - 1000 mL:** To prepare 1000 mL of gel destaining solution, add the following in a beaker and mix.

- Milli-Q or purified water - 450 mL
- Glacial acetic acid - 100 mL
- Methanol - 450 mL

- **Preparation of 5X Sample solubilizing buffer (SSB) - 100 mL**

The composition of 5X SSB to be prepared in milli-Q or purified water is as follows:

- 1 M Tris-HCl, pH 6.8 - 25 mL
- SDS - 5 g
- Glycerol - 30 mL
- Bromophenol blue - 20 mg

Dissolve all the above contents in 30 mL milli-Q or purified water and make up the volume to 95 mL with milli-Q or purified water. Once the contents are dissolved aliquot 950 µL of mixture in 1.5 mL of micro centrifuge tube. Store the aliquot at room temperature.

- **Preparation Solution B [3x Gel Buffer, pH 8.45] – 50 mL:** Add 18.16 g of Tris, 0.75 mL of 20% SDS and 10 mL of 5 N HCl or 4.31 mL of 37% HCl mix the components by using a magnetic stirrer, adjust the pH to 8.45 and make up the volume to 50 mL with milli-Q or purified water. Store at room temperature.
- **Preparation of 1X Cathode Buffer, pH 8.25 - 1000 mL:** Add 12.11 g of Tris, 17.92 g of Tricine and 5 mL of 20 % SDS or 1 g of SDS, mix the components by using a magnetic stirrer, adjust the pH to 8 and make up the volume to 1000 mL with milli-Q or purified water. Store at room temperature.
- **Preparation of 1X Anode Buffer, pH 8.9 -1000 mL:** Add 12.11 g of Tris, 3 mL of 5 N HCl, mix the components by using a magnetic stirrer, adjust the pH to 8.9 and make up the volume to 1000 mL with milli-Q or purified water. Store at room temperature.
- **Preparation of solution A (49.5 %) - 50 mL:** Weigh 24.0 g of acrylamide and 0.75 g of bisacrylamide, dissolve in 30 mL of milli-Q or purified water using a magnetic stirrer and make up the volume to 50 mL with milli-Q or purified water.

  Note: While preparing, avoid exposing the solution to light for an extended period.

Calculation for sample

Table 3: Calculation for sample				
Stock concentration (µg/µL)	**Required quantity (µg)**	**Required concentration (µg/ µL)**	**Volume of protein (µL)**	**Total volume for reaction (µL)**
15.73	50	1.0	3.18	50

Dilution for enterokinase enzyme

Table 4: Dilution for enterokinase enzyme				
Serial no.	**Dilutions**	**Volume of stock (µL)**	**Volume of buffer (µL)**	**Total volume (µL)**
1.	1:10	50	450	500
2.	1:100	5	495	
3.	1:1000	5	495	
4.	1:10000	5	495	

Unit definition: One unit of enterokinase enzyme is defined as the amount of enzyme that will cleave 50 µg of fusion protein in 16 hours at 22°C, in a buffer containing 20 mM Tris-HCl pH 7.4, 50 mM NaCl, 2 mM $CaCl_2$.

4. Protocol:

Day 01

- Make dilutions of enterokinase in EK dilution/storage buffer.
- Set up reactions, including a reaction without EK for the control.
- 50 µg of fusion protein in a buffer containing 20mM tris-HCl pH 7.4, 50 mM NaCl, 2mM $CaCl_2$.

Table 5: Reaction mix	
Component	**Reaction mix (µL)**
Protein	3.18
EK cleavage/Capture buffer	5
Diluted EK	2
Deionized water	33
Total volume	50

- Note: 2 µL of EK dilution/storage buffer for control.
- Mix well and incubate at 22 °C for 16 hours.

Day 02:

Preparation of SDS-PAGE (Tricine gel)

- Prepare the SDS tricine gel in 50 mL centrifuge tube for 1 mm notch plate as mentioned in table below according

Table 6: Composition for Tricine gel			
Components	Separating gel	Spacer gel	Stacking gel
Water	2.72 mL	2.72 mL	4.2 mL
Solution A	4 mL	1.22 mL	500 µL
Solution B	4 mL	2 mL	1.55 mL
Glycerol	1.28 mL	-	-
14% APS	40 µL	20 µL	80 µL
TEMED	15 µL	6 µL	4 µL

- Fill 1X cathode buffer inside the gel cassette (in between plates) and fill 1X anode buffer in the gel tank for tricine gel.
- Ensure that the buffer does not leak from base of the holder. If it does, reassemble the plates with minimal gap at the base.
- Gently remove the combs and flush each well with small amounts of the gel running buffer using a pipette.
- Load 10 µL of reaction mix tris tricine gel to determine EK digestion of protein.
- Insert the power plugs in respective color-coded sockets provided on the power pack (red/black).
- Run the gel at 70 V for Tricine gel.
- Run the gel until the blue dye front reaches the bottom end of the gel or until the visible marker bands are separated well enough.
- Once the gel run is complete, remove the plates from the holder and separate them carefully. Ease the separation of gel from the plates by washing the plates with water. Collect the gel using the gel remover or by inverting the plate into a container containing water to prevent gel from tearing.

Table 7: Gel specifications		
Serial no.	**Specification**	**Parameter for in-house gel**
1.	Gel	Tris-tricine
2.	Running buffer	1X Cathode buffer 1X Anode buffer
3.	Loading volume	10 µL
4.	Voltage	70V
5.	Marker	SeeBlue Plus2 Prestained standard

Staining of the gel

Coomassie Staining and Destaining

- Stain the gel using Coomassie brilliant blue stain.
- Post-staining, discard the Coomassie solution in an organic solvent discard container.
- Destain the gel using de-staining solution. The destaining process should be carried out until the desired contrast is obtained.

- Wash the gel with Milli Q water to remove residual stain and increase the protein band-gel contrast.
- Capture the image using Chemidoc Imaging system.